MANOGNA SISTA

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SUMMARY

Computer Science (AI & ML) undergraduate with a 9.35/10 CGPA and strong foundations in Data Structures & Algorithms, Object-Oriented Programming, DBMS, and Software Engineering. Proficient in Python, Java, C, SQL, JavaScript, and REST API development, with hands-on experience building scalable applications using Flask and MongoDB. Solved 700+ DSA problems across competitive programming platforms, strengthening algorithm design, complexity analysis, and problem-solving skills. Experienced in Agile team environments with strong technical communication, debugging, and software development skills.

EDUCATION

VNR Vignana Jyothi Institute of Engineering and Technology, Hyderabad, India

Bachelor of Technology in Computer Science Engineering (AI & ML); CGPA: 9.39/10 **Aug 2024 – May 2028**

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Machine Learning, Software Development Life Cycle (SDLC)

TECHNICAL SKILLS

Programming Languages: Python | Java | C | SQL | JavaScript

Data Structures & CS Fundamentals: Data Structures & Algorithms | Object-Oriented Programming | Complexity Analysis | DBMS | Operating Systems | Computer Networks | SDLC | Debugging |CI/CD pipelines.

Web & API Development: HTML | CSS | Flask | REST APIs | Web Services

Databases: SQL | MongoDB

Developer Tools: Git | GitHub | VS Code

PROJECTS - GitHub

Agriculture RAG Assistant — Evidence-Grounded Farming Guidance | [Python, Flask, SentenceTransformers, FAISS, OpenAI API]

- Developing a Retrieval-Augmented Generation assistant that retrieves relevant passages from curated agricultural documents to support fertilizer and soil-related guidance.
- Planned a semantic-search pipeline using SentenceTransformers embeddings and FAISS retrieval, with an LLM generating explanations grounded in retrieved sources.
- Designed as an explanation layer for the Fertilizer Usage Optimizer, connecting recommendations with supporting agricultural knowledge and references.

Digital Twin — AI-Based AQI Forecasting System | [Python, LSTM, Flask, REST APIs, Leaflet.js]

- Engineered a real-time AQI forecasting application using Python and LSTM, processing and cleaning 8,700+ hourly records into a unified time-series dataset.
- Integrated live AQICN/CPCB API data with historical records to support location -aware predictions across multiple monitoring stations.
- Designed and implemented REST APIs powering a Flask + Leaflet.js application for interactive AQI visualization and historical trend analysis.
- Optimized the end-to-end data pipeline and backend processing to reduce prediction latency and deliver responsive results to end users.

Toxic Comment Classifier — Multi-Label NLP-Based Toxicity Detection[Python,BERT, TF-IDF, LIME, Scikit-learn, Logistic Regression, FastAPI, Gradio]

- Developed an end-to-end NLP application leveraging Toxic-BERT and TF-IDF-based Logistic Regression to identify six categories of toxic content: toxic, severe toxic, obscene, threat, insult, and identity hate.
- Implemented text tokenization, feature extraction, and category-specific thresholding to enable multi-label classification based on model-generated prediction probabilities.
- Integrated the classification pipeline with a FastAPI backend and Gradio interface, enabling interactive comment analysis and structured toxicity prediction results.

Faculty Connect — Faculty Locator Web Application | [HTML, CSS, Flask, MongoDB, REST APIs, Git]

- Developed a full-stack faculty management application using Flask, MongoDB, HTML, CSS, and REST APIs to enable real-time faculty availability and search across departments.
- Designed and implemented RESTful APIs for authentication, faculty information management, and search, enabling efficient communication between frontend and backend services.
- Collaborated in a 5-member Agile team using Git and GitHub for version control, code reviews, debugging, and sprint-based development.
- Debugged API workflows and optimized MongoDB queries and backend logic to improve data retrieval efficiency, application re liability, and performance.
- Developed and executed API test cases to validate authentication, search, and faculty -management workflows, identifying and resolving backend issues.

Fertilizer Usage Optimizer — ML-Based Recommendation Engine | [Python, OOP, Machine Learning]

- Built an AI-based fertilizer recommendation system in Python, analysing NPK soil-nutrient values across multiple soil types to predict optimal fertilizer requirements.
- Achieved ~80% recommendation accuracy on validated agricultural datasets through iterative model tuning.
- Architected the application using Object-Oriented Programming principles with a modular, scalable design supporting both manual NPK input and soil-type-based prediction modes.

ACHIEVEMENTS

- Solved **700+ Data Structures & Algorithms problems** across LeetCode, CodeChef, HackerRank, and NeetCode, strengthening algorithm design, complexity analysis, and problem-solving skills.
- **State-Level Finalist, IDEATHON** (Government of Telangana — DEET & Gurthimpu Foundation): selected among 1,500+ participants for AGRI-VEDA, an AI/ML-based agriculture innovation project.
- Maintained an **80+ day NeetCode streak**, consistently practicing interview-oriented Data Structures & Algorithms.
- **Lead RJ, VJ Vibes College Podcast Club:** Led a team of 50+ members, coordinating planning, execution, and delivery of 60+ podcast sessions while collaborating across multiple student organizations.
- **Google Developer Groups on Campus (GDGC) – Women in Tech:** Participated in 18+ technical workshops and industry-led sessions focused on emerging technologies.
- **Ignition Wing Master, VJ STARTUPS:** Leading a student startup and innovation community focused on developing new ideas and entrepreneurial

initiatives with a team of 40 people.

- **Blog Writer, Medium:** Authored 10+ technical and reflective articles, simplifying complex concepts for a wider audience.